

Lab: system calls

Sandbox a command(moderate)

Output

```
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 2 starting
hart 1 starting
init: starting sh
$ sandbox 32768 - cat README
cat: cannot open README
$ |
```

Sandbox Grade test:

```
FatimaFaisal:~$ ./grade-lab-syscall sandbox_mask
make: 'kernel/kernel' is up to date.
== Test sandbox_mask == sandbox_mask: OK (0.9s)
FatimaFaisal:~$ |
```

Sandbox with allowed pathnames(easy)

Output

```
FatimaFaisal:~$ make qemu
mv -f fs.img fs.img.bk
mkfs/mkfs fs.img README user/exec.sh user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_sandbox user/_attack user/_secret
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
balloc: first 1027 blocks have been allocated
balloc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ sandbox 32768 README grep xv6 README
xv6 is a re-implementation of Dennis Ritchie's and Ken Thompson's Unix
Version 6 (v6). xv6 loosely follows the structure and style of v6,
xv6 is inspired by John Lions's Commentary on UNIX 6th Edition (Peer
(Kaashoek,rtm@mit.edu). The main purpose of xv6 is as a teaching
$ |
```

Attack xv6 (moderate)

Output

```

printf.o user/umalloc.o
riscv64-unknown-elf-objdump -S user/_attack > user/attack.asm
riscv64-unknown-elf-objdump -t user/_attack | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/attack.sym
riscv64-unknown-elf-gcc -Wall -Werror -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -DSOL_SYSCALL -DLAB_SYSCALL -MD -mcode
l=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin
-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-ex
it -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-buili
n-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o user/secret.o user/secret.c
riscv64-unknown-elf-ld -z max-page-size=4096 -T user/user.ld -o user/_secret user/secret.o user/ulib.o user/usys.o user/
printf.o user/umalloc.o
riscv64-unknown-elf-objdump -S user/_secret > user/secret.asm
riscv64-unknown-elf-objdump -t user/_secret | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/secret.sym
mkfs/mkfs fs.img README user/exec.sh user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/
_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/
_forphan user/_dorphan user/_sandbox user/_attack user/_secret
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
ballocc: first 1033 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-l
egacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 2 starting
hart 1 starting
init: starting sh
$ secret xyzzy
$ attack
xyzzy
$ |

```

Attack Grade test

```

FatimaFaisal:~$ cd xv6-labs-2025
FatimaFaisal:~$ ./grade-lab-syscall attack
make: 'kernel/kernel' is up to date.
== Test attack == attack: OK (1.0s)
      (Old xv6.out.attack failure log removed)
FatimaFaisal:~$ ./grade-lab-syscall attack
make: 'kernel/kernel' is up to date.
== Test attack == attack: OK (1.1s)
FatimaFaisal:~$ |

```

Make grade:

Both of the text files are uploaded

```

riscv64-unknown-elf-objdump -S kernel/kernel > kernel/kernel.asm
riscv64-unknown-elf-objdump -t kernel/kernel | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > kernel/kernel.sym
make[1]: Leaving directory '/home/maskatia/xv6-labs-2025'
== Test answers-syscall.txt ==
answers-syscall.txt: FAIL
      Cannot read answers-syscall.txt
== Test sandbox_mask ==
$ make qemu-gdb
sandbox_mask: OK (3.9s)
== Test sandbox_fork ==
$ make qemu-gdb
sandbox_fork: OK (0.8s)
== Test sandbox_path ==
$ make qemu-gdb
sandbox_path: OK (1.3s)
== Test sandbox_most ==
$ make qemu-gdb
sandbox_most: OK (0.7s)
== Test sandbox_minus ==
$ make qemu-gdb
sandbox_minus: OK (1.1s)
== Test attack ==
$ make qemu-gdb
attack: OK (1.2s)
== Test time ==
time: FAIL
      Cannot read time.txt
Score: 39/45
make: *** [Makefile:369: grade] Error 1

```